Weather, Interconnections, and Energy Prices

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Abstract

Across five seeds and 49 market identities, a pooled MLP scores 21.19 EUR/MWh test MAE versus 20.97 for real-graph GAT. Self-loop GAT and pooled XGBoost each score 21.06. Graph gains are smaller after controlling for pooled learning. Feature effects are mixed and GRU histories worsen scores. These are conditional exploratory results with unresolved publication vintages.

1 Approach

At 12:00 Europe/Brussels on calendar day $D-1$, predict every hourly day-ahead price of delivery day D, including 23/25-hour daylight-saving days. We implement an independent dataset builder, per-zone XGBoost [1], and pooled hourly GATv2 models [2]. Directed graph edges are physical pairs observed during training; historical node lifetimes mask inactive identities. A shuffled control rewires non-self edges, preserving static in/out degrees without duplicates. All GAT variants add self-loops, testing real topology under the same neural learner.

2 Experiments

Exact data and partitions. The master contains 5,887,954 zone-hours for 62 price-area identities, January 2016–October 2026. ENTSO-E prices, load forecasts and physical flows use releases v1.2. Additional sources are PECD/ERA5 temperature, anomaly and wind/solar capacity factors (2010–September 2026), and original CWE/CORE JAO exports (May 2015–October 2026). Weather is reanalysis, not archived forecasts; its requested v1.1 release identity is unverified. Missing values and negative prices are retained, and DE_AT_LU remains separate from DE_LU. Forty-nine identities have valid targets in fit, validation and test; the other 13 remain explicitly unscored, while all 62 can provide dated graph context.

Fit on January 1, 2016–March 31, 2024 (pretrain + train); validate on April 1, 2024–March 31, 2025; test on April 1, 2025–September 30, 2026, using Brussels delivery dates. October stays unassigned. MAE and RMSE are measured after inverse transformation, in EUR/MWh. A 1 MWh quantity is 1,000 kWh; 21 EUR/MWh error is 2.1 euro cents/kWh. Relative MAE divides model MAE by weekly-persistence MAE on identical hours; 0.72 means 28% less error, not 28% price error. Weekly persistence copies the same identity's price 168 elapsed hours earlier. Targets are never filled or filtered for feature completeness; neural preprocessing is fitted on training-period features only.

Controlled models. The 100 jobs cross five seeds (224–228): per-zone/pooled trees, MLP/GAT feature ablations, self-loop and three degree-preserving rewired graphs, and GRU–MLP/GAT pairs. Rich static MLP/GAT have 12,001/12,097 parameters; neither uses zone embeddings. Neural models share training-only preprocessing, asinh(price/50), L1 loss, Adam 0.001, maximum 30 epochs and validation patience five. GAT has two layers, 64 channels and four heads. The 32-unit GRU encodes 168 same-identity completed price intervals ending at Brussels noon, with missingness. Trees use depth 6 and at most 600 rounds.

Multi-seed comparison. The richer-input controls use 428,213 validation and 633,885 test zone-hours with valid targets and finite weekly history. Values are mean $\pm$ sample standard deviation across five seeds, not prediction ensembles. Other feature sets, per-zone/country results, training-defined price-tail/season slices and rewiring variability are in the generated study appendix.

Forecasting method	Validation MAE	Test MAE	Test rMAE
Weekly persistence	30.18	29.13	1.000
Pooled XGBoost	**20.74 ± 0.01**	21.06 ± 0.01	0.723
Pooled MLP	21.44 ± 0.23	21.19 ± 0.21	0.727
Real-graph GAT	20.98 ± 0.11	**20.97 ± 0.13**	**0.720**
Self-loop GAT	21.27 ± 0.09	21.06 ± 0.08	0.723
Shuffled GAT (3 graphs)	21.21 ± 0.05	21.45 ± 0.07	0.736
GRU + MLP	24.37 ± 1.65	24.50 ± 1.22	0.841
GRU + real GAT	23.62 ± 0.52	23.59 ± 0.51	0.810

Table 1: Grouped by model class; learned models use richer inputs. MAE in EUR/MWh; rMAE divides test MAE by weekly-persistence MAE on the same hours (1 = baseline). Bold denotes the smallest mean per column, not statistical significance.

Graph contribution. Real GAT improves test MAE by 0.21 (1.00%) against MLP, but only 0.09 against self-loop GAT and 0.08 against pooled XGBoost. Pooled XGBoost has the best validation score among richer-input controls; the small test advantage does not justify selecting GAT retrospectively. MLP-to-GAT validation improvement is 0.46 (95% paired week interval [0.20, 0.86]). Weeks are shared across countries, conditional on fitted models. Real GAT beats all three shuffled means on test; one shuffled-control validation interval includes zero.

Earlier and extended experiments. The first single-seed real/shuffled GAT scores of 20.83/21.54 become 20.97/21.45 across seeds and rewirings. Pooled rich XGBoost scores 21.06 versus 22.11 for per-zone models. Gains vary: France GAT scores 25.48 versus MLP's 22.00. Weather-only GAT has test MAE 20.90, but richer GAT has better validation MAE (20.98 versus 21.06). Load and JAO effects are mixed. GRU–MLP/GAT score 24.50/23.59, worse than static controls; these settings do not rule out other temporal models or detailed constraints.

Timing and limitations. Prior-day auction prices are normally published on $D - 2$ [3]; DST guards exclude lag-24 target-day prices. ERA5-derived lag-48 inputs conflict with documented publication latency [4]; load/JAO vintages remain unresolved. Temporal histories end by noon. Static training-only topology does not reconstruct commissioning dates. All models retain identical target keys/masks and missing features; targets are never filled. Previously inspected test data make these conditional exploratory comparisons, not confirmatory evidence. Multiple feature comparisons do not establish causal effects.

3 Future work

Build a daily forecast service that saves predictions before Brussels noon, together with received inputs and model versions. Score forecasts when auction results arrive; monitor rolling rMAE, data completeness, deadline misses and fallback use. Start with price-only pooled XGBoost and persistence, then compare MLP/GAT and timely station observations. This prospective experiment measures accuracy, drift and operational reliability. Assess detailed constraint graphs after establishing the baseline service.

References

[1] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. *KDD*, 2016. arXiv:1603.02754.

[2] S. Brody, U. Alon and E. Yahav. How attentive are graph attention networks? *ICLR*, 2022. arXiv:2105.14491.

[3] Nord Pool. Day-ahead auction timeline. Accessed November 2026.

[4] ECMWF. ERA5 global atmospheric reanalysis. Accessed November 2026.